Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Equilibrium

Meaning in my words How would you explain equilibrium to a friend?	**How can you tell?** A cart rolls straight at steady speed. What must the horizontal forces add up to?
Example or sketch Sketch the horizontal forces on a cart with a 3 N pull right and 3 N friction left.	**Non-example and why** A tossed ball is briefly still at the top. Is it in equilibrium there?

Balance

Meaning in my words How would you explain balance to a friend?	**How can you tell?** Two cables suspend a sign at rest. What must their upward components add up to?
Example or sketch A sign weighs 12 N and two cables share the upward support equally. What is each vertical contribution?	**Non-example and why** The horizontal components cancel. Is that enough to show equilibrium?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Symmetry

Meaning in my words How would you explain symmetry to a friend?	**How can you tell?** Two identical cables hold a centered sign at equal angles. What tells you their tensions match?
Example or sketch Sketch equal tension arrows. What happens to the two horizontal components?	**Non-example and why** One cable is much steeper. Can you still assume the tensions are equal from symmetry alone?

Rest

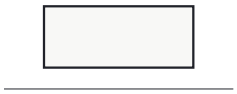
Meaning in my words What does it mean for an object to be at rest at an instant?	**How can you tell?** At the top of a ball's straight upward toss, what is zero and what is still acting?
Example or sketch How would you draw velocity and acceleration at that instant?	**Non-example and why** "It stopped, so the forces must balance." What does the next instant show?

Doodle the idea

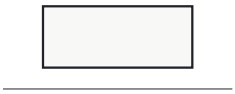
SECTION B / DRAW WITH YOUR TEACHER Name _____

Sketch two systems with balanced forces: one at rest and one moving steadily. Label the common acceleration.

At rest



Moving steadily



My marks mean

On assignment: Equilibrium

SECTION E / BACK / FINAL MEETING ONLY Name _____

The flight display reads “net force zero.” XJ’s last velocity was +3 km/s. Two graphics say STOPPED and COASTING.

PICK A JOB / DELIVER ONE PRODUCT

☐ **Flight controller**

Select the status graphic. Plot a second position two seconds later relative to the first and explain your instruction.

☐ **Broadcast producer**

Choose the lower-third for a live shot. Ask the anchor to distinguish zero acceleration from zero speed before airing it.

One partner makes the graphic; one delivers the first call. Show your evidence to the crew or the audience.

OUR GRAPHIC / COMMAND OR ON-AIR CUE

INCOMING UPDATE / WAIT FOR YOUR TEACHER

Listen to the new report. Switch speakers. Keep or change your call; point to the new evidence.

Editor’s check: Check the claimed balance along both coordinate axes.